CLASS ACTIVITY — Solutions & Answers

Data Visualization & Statistical Analysis Case Studies

1. Case Study: Monthly Hospital Patient Visits

A hospital administrator wants to analyze the number of patients visiting the hospital each month during 2025.

Patients

Month	Jan	Feb	Mar	Apr	May	Jun
1200	1350	1500	1450	1700	1850	

1. Which visualization is most suitable?

> **Line Chart** or **Bar Chart**. Line charts are best suited for illustrating trends over continuous time intervals.

2. What trend can be observed?

> An overall **upward** / **increasing trend** in patient visits (rising from 1,200 in January to 1,850 in June, despite a minor drop in April).

3. Identify the month with the highest patient count.

> **June** (with 1,850 patients).

4. Predict whether visits are increasing or decreasing.

> Visits are **increasing**.

2. Case Study: Study Hours vs Exam Marks

A teacher wants to determine whether studying more hours improves exam performance.

Student	Study Hours	Marks
A	2	45
B	3	52
C	4	58
D	5	68
E	7	82

1. Which chart is appropriate?

> A **Scatter Plot**.

2. Is there a positive or negative relationship?

> **Positive relationship** (as study hours increase, exam marks increase accordingly).

3. Which student is an outlier, if any?

> **None**. All data points follow a consistent positive trend line.

4. Can marks be predicted from study hours?

> **Yes**, marks can be predicted reliably due to the strong positive linear correlation.

3. Case Study: Employee Salary Distribution

A company wants to understand how salaries are distributed among employees.

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Salary Data (in Thousands): 25, 28, 30, 32, 35, 35, 36, 38, 40, 42, 45, 48, 50, 52, 55, 60, 65, 70
```

1. Which graph should be used?

> A **Histogram** or **Box Plot**.

2. In which salary range do most employees fall?

> The **30k–40k range** (contains 7 employees).

3. Is the distribution symmetric or skewed?

> **Right-skewed (positively skewed)**, with most values clustered in the 25k–50k range and a tail extending to 70k.

4. Are there any salary ranges with very few employees?

> **Yes**, higher ranges like **60k–70k** have very few employees (only 3 employees).

4. Case Study: Favorite Programming Language

A survey was conducted among 200 students.

Students

Language	Python	Java	C++	JavaScript	R
80	50	30	25	15	

1. Which visualization is suitable?

> A **Bar Chart** or **Pie Chart**.

2. Which language is most popular?

> **Python** (80 students / 40%).

3. Which language is least preferred?

> **R** (15 students / 7.5%).

4. Compare Python and Java popularity.

> Python is **1.6 times more popular** than Java (80 vs. 50 students; 40% vs. 25%).

5. Case Study: Household Monthly Expenses

A family spends money in the following categories:

Expense (%)

Category	Rent	Food	Education	Transportation	Entertainment
40%	25%	15%	10%	10%	

1. Which chart best represents the data?

> A **Pie Chart** or **Donut Chart** (ideal for displaying proportions of a whole summing to 100%).

2. Which category consumes the largest budget?

> **Rent** (40%).

3. What percentage is spent on Food and Education together?

> **40%** (25% Food + 15% Education).

4. Which categories occupy the smallest share?

> **Transportation** and **Entertainment** (10% each).

6. Case Study: Student Test Scores

Scores obtained by students:

```
Scores: 45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95
```

1. Which plot should be used?

A **Box Plot** (Box-and-Whisker plot) or **Stem-and-Leaf plot**.

2. What is the median score?

65 (the 8th value in the sorted set of 15 scores).

3. Are there any outliers?

Yes, 95 can be considered an upper outlier relative to the main concentration of scores.

4. What is the spread of the data?

- **Range:** 50 (95 - 45)
- **Interquartile Range (IQR):** 20 ($Q_3 = 75$, $Q_1 = 55$; 75 - 55 = 20)